Fiscal Year:	FY 2016	**Task Last Updated:**	FY 04/06/2016
PI Name:	Salas, Eduardo Ph.D.		
Project Title:	Dynamic Team Role Allocation in Long Duration, Exploration Missions: Identification of Roles, Triggers, and Measurement Tools		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	eduardo.salas@rice.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-348-3917
Organization Name:	Rice University		
PI Address 1:	Department of Psychology		
PI Address 2:	6100 Main Street MS25		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77005	**Congressional District:**	7
Comments:	NOTE: Previous affiliation was University of Central Florida, until mid-2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	07/03/2014	**End Date:**	11/02/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
Contact Email:	lauren.b.leveton@nasa5.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed per PI move to Rice University in fall 2015; information per D. Risin/JSC (Ed., 4/5/16)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Burke, Shawn Ph.D. (University of Central Florida) Driskell, James Ph.D. (Florida Maxima Corp.) Fiore, Stephen Ph.D. (University of Central Florida)		
Grant/Contract No.:	NNX14AM73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Long duration exploration missions present a unique environment characterized by many stressors (e.g., social isolation, danger, confinement, interpersonal dynamics, periods of over/under stimulation), with little ability to escape. Research has found that within such environments interpersonal dynamics occupy a key role in effective functioning (Forsyth, 2010). While the last few years has witnessed an increase in research examining the composition requirements of high performance teams, little work has examined these issues in light of teams embedded in long duration, exploration missions. Therefore, we describe a program of work which addresses Team Gaps 1, 4, and 8 in NASA's Human Research Roadmap. We seek to answer the following questions with regard to long duration, exploration missions: (1) what are the key social and team technical (task) roles which influence team function; (2) what are the behavioral and communicative markers which can be used to assess the degree to which key identified social and team technical roles are being fulfilled; (3) what contextual aspects serve to trigger a need for the dynamic shift of social roles; (4) what are the optimal combinations (i.e., profiles, algorithms) of social roles for the maintenance and regulation of team functions; (5) what are the markers that can be used to select for those most likely to fit social profiles and how do these profiles change across the duration of the mission (i.e., the team's life cycle). In answering these questions we seek to provide a series of scientifically grounded and experimentally validated taxonomies, guidelines, and measurement tools for team selection/composition. In exploring these questions, we take a multi-pronged approach consisting of analysis of archival data (e.g., astronaut diaries, historical accounts of teams operating in in isolated, confined environments (ICE), prior collected University of Central Florida astronaut interviews), scientific literatures on group dynamics, personality, team roles, stress, and diversity, interviews, and experimentation in NASA analogs. Forsyth D. R. (2010). Group dynamics. J. D. Hague (ED.). Belmont CA: Wadsworth, Cengage Learning.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research contributes to a better understanding of the task and social roles needed for teams to effectively function, especially within the context of isolated, confined environments such as long duration spaceflight.
Task Progress:	NOTE: Principal Investigator moved to Rice University in fall 2015. Project continues at Rice with the same title, period of performance 11/3/2015-11/2/2017, grant #NNX16AB08G. See that project for continued reports.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Articles in Peer-reviewed Journals	Salas E, Tannenbaum SI, Kozlowski SWJ, Miller CA, Mathieu JE, Vessey WB. "Teams in space exploration: A new frontier for the science of team effectiveness." Curr Dir Psychol Sci. 2015 Jun;24(3):200-7. http://dx.doi.org/10.1177/0963721414566448 , Jun-2015